

Practical 1: File Upload Validation Assessment

Objective

To assess the effectiveness of file upload validation mechanisms implemented within a web application and determine whether file type restrictions could be bypassed.

Tools Used

- Burp Suite
- Mozilla Firefox
- Internshala Ethical Hacking Practice Lab

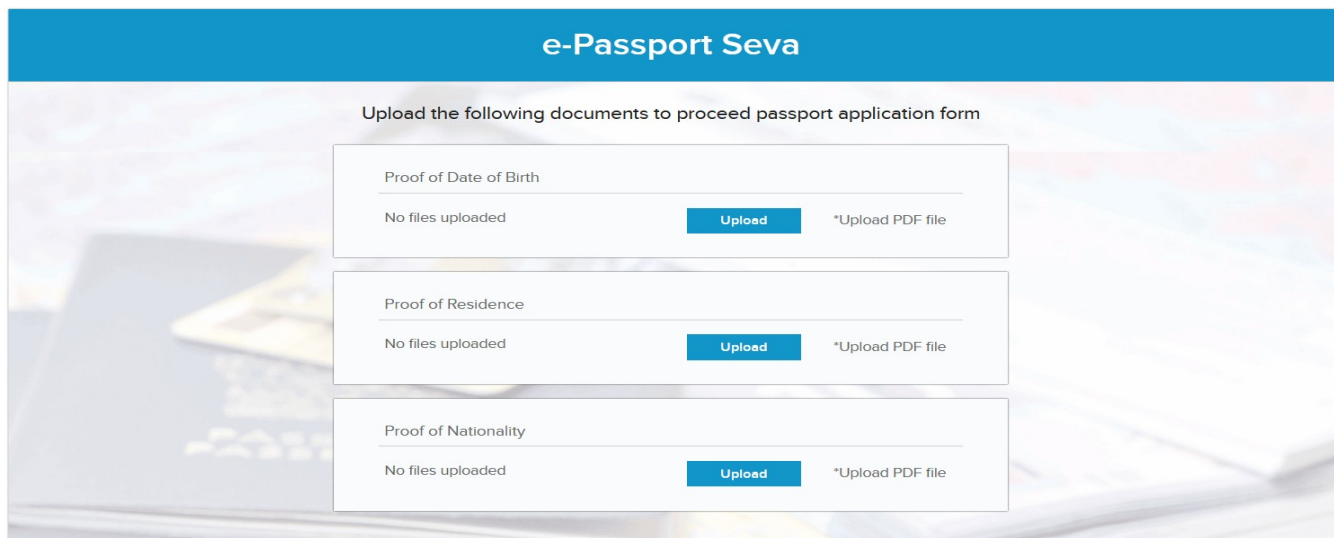
Target Application

The assessment was conducted against the file upload functionality provided within the Internshala Ethical Hacking Practice Lab. The application required users to upload PDF documents as part of a passport application process.

Methodology

Step 1: Application Analysis

The upload interface was reviewed to understand the application's requirements and restrictions. The application indicated that only PDF files were permitted for document submission.

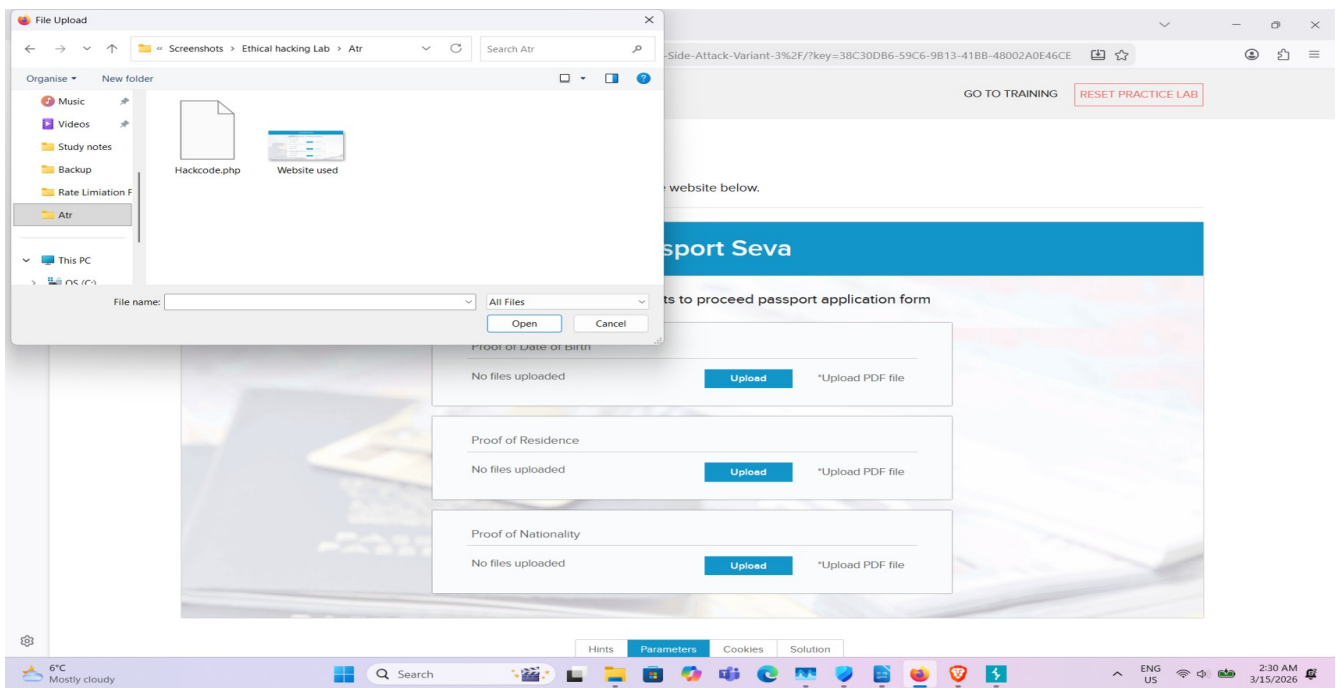


The screenshot displays the 'e-Passport Seva' web interface. At the top, a blue header contains the text 'e-Passport Seva'. Below the header, a message states: 'Upload the following documents to proceed passport application form'. The interface features three distinct upload sections, each with a title, a file input field, an 'Upload' button, and a note: '*Upload PDF file'.

Document Type	Status	Action	Requirement
Proof of Date of Birth	No files uploaded	Upload	*Upload PDF file
Proof of Residence	No files uploaded	Upload	*Upload PDF file
Proof of Nationality	No files uploaded	Upload	*Upload PDF file

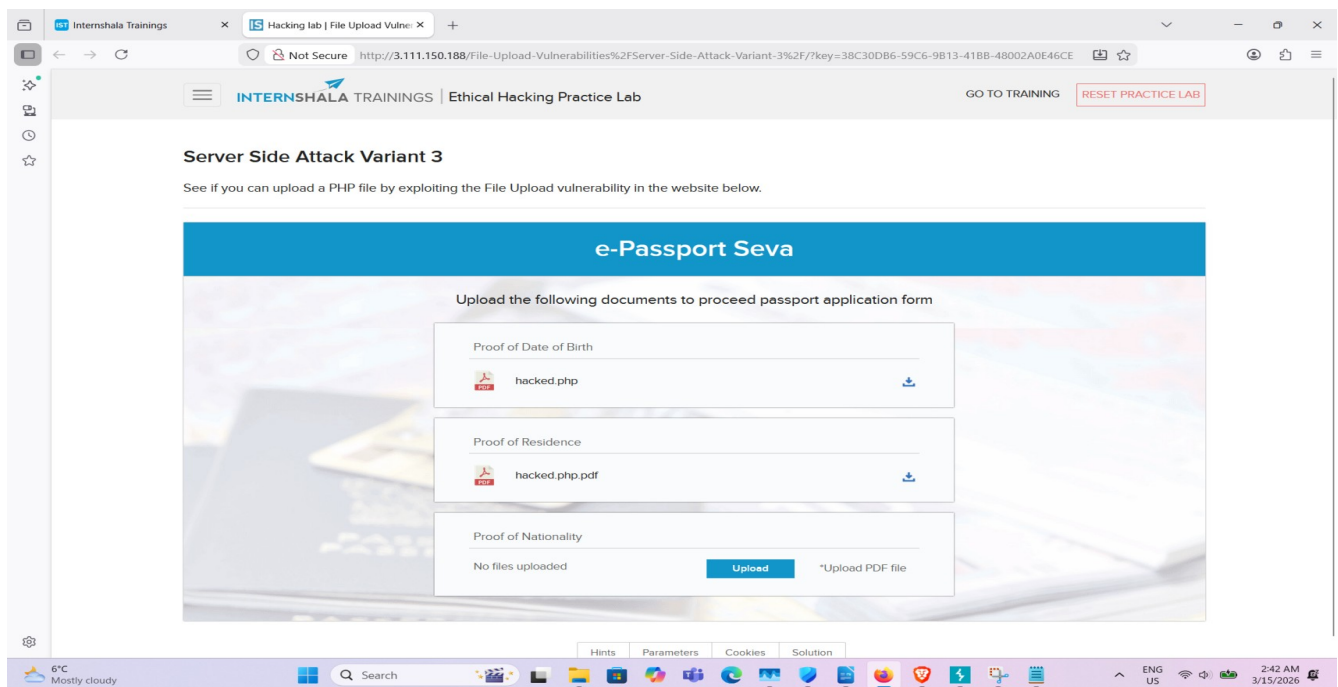
Step 2: Request Interception and Analysis

The file upload request was intercepted using Burp Suite. The intercepted request was examined to identify how the application validated uploaded files and what information was transmitted to the server during the upload process.



Step 3: Validation Testing

The uploaded file's filename and related request parameters were analyzed to determine whether validation relied solely on file extensions. Testing demonstrated that modifying the filename format allowed the application to process the upload despite the intended file-type restriction.



Result

The application accepted an uploaded file that did not strictly conform to the intended upload restrictions, indicating weaknesses in the server-side file validation process.

Security Impact

Weak file upload validation can expose applications to security risks if unauthorized file types are accepted and stored by the server. Proper validation should include extension verification, MIME type validation, content inspection, and secure storage controls.

Learning Outcome

This practical exercise provided hands-on experience in evaluating file upload security controls and understanding the importance of robust server-side validation mechanisms in web applications.